

The Complete One-Gap Spectrum of the Degree-One Lamé Hamiltonian

Route-A source-local certificate HCS-C340

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Abstract

We prove the complete real-line spectrum of the degree-one Lamé operator in a fixed Jacobi convention. Three explicit band-edge modes, a skew third-order commuting operator, and its cubic algebraic relation yield two bands, one finite open gap, and pure absolute continuity. The free-modulus and soliton limits, threshold resonance, translations, and all real-convention boundaries are closed explicitly. Finite exact receipts are separated from the infinite Floquet proof, and the source, collision, and Route-A boundaries are audited.

1 Operator convention and band-edge modes

Fix $m = k^2 \in (0, 1)$ and write $K = K(m)$. On $L^2(\mathbb{R})$ let

$$H_m = -D^2 + u(x), \quad D = \frac{d}{dx}, \quad u(x) = 2m \operatorname{sn}^2(x \mid m), \quad (1)$$

with domain $H^2(\mathbb{R})$. The real potential is smooth, bounded, and has period $2K$. Hence the stated realization is self-adjoint and bounded below. All Floquet multipliers below are measured over $2K$.

The Jacobi differential identities are

$$\operatorname{sn}'' = -(1+m)\operatorname{sn} + 2m\operatorname{sn}^3, \quad (2)$$

$$\operatorname{cn}'' = (2m\operatorname{sn}^2 - 1)\operatorname{cn}, \quad \operatorname{dn}'' = (2m\operatorname{sn}^2 - m)\operatorname{dn}. \quad (3)$$

Consequently

$$H_m \operatorname{dn} = m \operatorname{dn}, \quad H_m \operatorname{cn} = \operatorname{cn}, \quad H_m \operatorname{sn} = (1+m)\operatorname{sn}. \quad (4)$$

Over one potential period, dn is periodic while cn and sn are antiperiodic. These formulas identify three edges, but by themselves do not prove that all higher gaps close.

2 Commuting operator and spectral curve

Set

$$A = -4D^3 + \{6u - 4(1+m)\}D + 3u'. \quad (5)$$

For $\theta \in [-\pi, \pi)$ define

$$H_\theta^j = \{f \in H^j([0, 2K]) : f^{(\ell)}(2K) = e^{i\theta} f^{(\ell)}(0), \ 0 \leq \ell < j\}.$$

The closed fiber realization of (5) has

$$D(A_\theta) = H_\theta^3([0, 2K]). \quad (6)$$

The coefficient of D has derivative $6u'$. Since all coefficients are $2K$ -periodic, every endpoint product in the Green boundary form cancels: the common phase $e^{i\theta}$ cancels its conjugate. Three integrations by parts therefore show $A_\theta^* = -A_\theta$; equivalently, the adjoint boundary form imposes the same three quasi-periodic conditions. Direct use of (2) gives the two identities

$$u'' = 3u^2 - 4(1+m)u + 4m, \quad (7)$$

$$(u')^2 = 2u^3 - 4(1+m)u^2 + 8mu. \quad (8)$$

Differentiating (7) and expanding differential-operator products, with (8) for the zeroth-order remainder, proves

$$[A, H_m] = 0, \quad A^2 = -16(H_m - m)(H_m - 1)(H_m - 1 - m). \quad (9)$$

The first integral is essential here; the stationary second-order equation alone leaves an arbitrary integration constant and does not prove the second identity in (9).

Theorem 1 (complete Lamé spectrum). *For every $0 < m < 1$,*

$$\sigma(H_m) = [m, 1] \cup [1 + m, \infty), \quad (10)$$

and the spectrum is purely absolutely continuous. Thus $(1, 1 + m)$ is the only finite open gap and every higher folded-zone gap is closed.

Proof. Let

$$R(E) = (E - m)(E - 1)(E - 1 - m).$$

Floquet decomposition realizes H_m as the direct integral of self-adjoint quasi-periodic operators with domain H_θ^2 on $[0, 2K]$. The operator A_θ has domain (6) and is skew-adjoint there. A fiber eigenfunction is smooth by its second-order ODE, lies in H_θ^3 , and is therefore in $D(A_\theta)$. If $H_m\psi = E\psi$ in a fiber, then (9) gives

$$-\|A\psi\|^2 = -16R(E)\|\psi\|^2.$$

Therefore a spectral energy must satisfy $R(E) \geq 0$.

Conversely, suppose $R(E) > 0$. On the two-dimensional solution space of $H_m\psi = E\psi$, equation (9) gives distinct eigenvalues $\pm 4i\sqrt{R(E)}$ for A . Because A commutes with period monodromy, its one-dimensional eigenspaces are Bloch lines. Real coefficients make the two Bloch solutions complex conjugates. If one has multiplier λ , the other has multiplier $\bar{\lambda}$. Their Wronskian is nonzero and constant; translating it by $2K$ multiplies it by $\lambda\bar{\lambda}$. Hence $|\lambda| = 1$, placing E in the spectrum. The three roots of R follow by closure and agree with (4). The condition $R(E) \geq 0$ is exactly (10).

For a real one-dimensional periodic Schrödinger operator, Floquet fibers have analytic nonconstant band functions; the direct integral therefore has neither eigenvalues nor singular-continuous spectrum. Its spectrum is purely absolutely continuous [4]. Since the second allowed interval extends from $1 + m$ to infinity without another excluded chamber, all higher folded-zone contacts are closed. This is an analytic completeness proof, not a finite coefficient inference. $\square$

The argument through (9) is the *commuting spectral-curve owner* of revision round zero.

3 Endpoint and equivalence atlas

At $m = 0$, equation (1) becomes the free operator $-D^2$ with spectrum $[0, \infty)$; the finite gap has closed. This face also explains why the infinitely many folded free contacts are not open gaps.

As $m \uparrow 1$, one has $K(m) \rightarrow \infty$ and, locally on the line,

$$u(x) \longrightarrow 2 \tanh^2 x = 2 - 2 \operatorname{sech}^2 x.$$

The lower band $[m, 1]$ collapses to the L^2 level 1, and the upper band tends to $[2, \infty)$. For the unshifted potential $-2 \operatorname{sech}^2 x$, the zero solution is a threshold resonance, not an additional L^2 eigenstate; after adding 2 it sits at threshold 2.

Changing k to $-k$ leaves $m = k^2$ and the operator unchanged. A spatial translation is unitary conjugacy and changes no band. Complex m , or real $m \notin [0, 1]$, lies outside the frozen real self-adjoint Jacobi convention. These statements are the *modulus boundary owner* added in revision round one.

4 Finite receipts and Route-A boundary

The executable ledger contains all 199 reduced rational moduli p/q with $0 < p < q \leq 25$. It stores exact edges, gap widths, cubic coefficients, sign chambers, and fiber labels. The checker reconstructs each fraction and each root. An independent SymPy lane composes (5) with (1) and reduces every coefficient using both (7) and (8). These receipts catch sign and convention errors; only the Floquet/Wronskian proof establishes every real energy and every higher gap.

The nearest workspace owners are different: C262 treats a piecewise-constant square-wave Hill transfer matrix; C327 treats a singular Kronig–Penney delta comb; C221 and C231 use nonperiodic soliton/front Hessians. None owns the smooth elliptic one-gap theorem.

The conservative Route-A tuple is

$$(A0_FAIL, A1_WEAK, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION).$$

The source-native self-adjoint Hamiltonian and its unitary time evolution support A4. They do not create rational-prime data, an arithmetic orbit dictionary, a target Euler product, root number, automorphy, target divisor or functional equation, target zero match, or a Hilbert–Polya operator. Route B remains false. This is the *finite evidence and route firewall* of the final revision.

Source boundary

This paper is a source-local reconstruction, not a priority claim. Ince’s papers own the periodic Lamé analysis; NIST DLMF supplies an authoritative identity index; Kuchment supplies the general Floquet direct-integral and periodic absolutely-continuous spectral theorem. The checked JSON grid and release gates are new implementation receipts and are not attributed to those sources.

References

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- [4] P. Kuchment, “An overview of periodic elliptic operators,” *Bulletin of the American Mathematical Society* 53 (2016), 343–414, DOI 10.1090/bull/1528.